

A Comparative Study of Cost-Sensitive Learning vs. Data Balancing in Credit Risk Modeling

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Abstract

This study presents a comparative analysis of cost-sensitive learning and data balancing techniques in the context of credit risk modeling. As financial institutions strive to improve the accuracy of credit assessments, the challenges posed by imbalanced datasets and the costs associated with misclassification become increasingly significant. This research evaluates the effectiveness of cost-sensitive learning approaches, which assign different costs to misclassifications, against traditional data balancing methods such as oversampling and undersampling. By applying these techniques to a real-world credit risk dataset, we assess their impacts on model performance metrics, including accuracy, precision, recall, and cost-effectiveness. Findings reveal that while both strategies can enhance predictive performance, cost-sensitive learning often provides a more nuanced approach that effectively addresses the financial implications of misclassification, leading to better-informed lending decisions.

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I. Introduction

A. Background on Credit Risk Modeling

Credit risk modeling is a fundamental task in finance and banking, designed to estimate the likelihood that a borrower will default on a loan or fail to meet contractual debt obligations. These models support key decisions across the **credit lifecycle**—from loan origination and pricing to portfolio management and regulatory capital allocation. Traditionally, statistical methods such as **logistic regression** and **scorecards** have dominated this space. However, the growing availability of granular borrower data and the advancement of machine learning (ML) algorithms have significantly enhanced the ability to predict credit defaults with higher precision and granularity.

In modern finance, credit risk models are expected to do more than just predict default—they must be **scalable**, **adaptable to changing market conditions**, and **sensitive to regulatory compliance**, including anti-discrimination laws.

B. Importance of Accuracy in Credit Assessments

Accurate credit assessments are critical to the financial system's health and stability. Poor predictions can lead to:

- **Underestimation of risk**, resulting in high default rates and financial losses.
- **Overestimation of risk**, which restricts access to credit for creditworthy borrowers, reducing potential revenue and contributing to financial exclusion.

Inaccuracies are particularly harmful in **highly imbalanced settings**, where the default class (minority) is underrepresented. A model may achieve high overall accuracy while **failing to detect the minority (high-risk) class**, undermining its practical utility. Therefore, increasing the **sensitivity and precision** for default predictions is as important as overall accuracy.

C. Objectives of the Study

This study seeks to:

1. Investigate the **impact of data imbalance** on the accuracy and reliability of credit risk models.
 2. Evaluate and compare the effectiveness of **data-level approaches** (oversampling and undersampling) and **algorithm-level approaches** (cost-sensitive learning).
 3. Design and implement **cost-sensitive machine learning models** that incorporate the **unequal misclassification costs** inherent in credit decision-making.
 4. Provide empirical insights into which approach or combination offers **optimal trade-offs between accuracy, recall, and financial impact**.
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II. Literature Review

A. Overview of Credit Risk Assessment Methods

Traditional credit scoring methods include:

- **Logistic Regression**: Common for its interpretability and regulatory acceptance.
- **Discriminant Analysis**: Separates borrower classes based on linear combinations of features.
- **Scorecards**: Rule-based systems assigning weights to features, often used by credit bureaus.

With the emergence of machine learning:

- **Decision Trees, Random Forests, Gradient Boosting (e.g., XGBoost)**: These offer greater flexibility and non-linearity handling.
- **Neural Networks**: Applied in high-dimensional data environments, though less interpretable.

- **Support Vector Machines (SVMs):** Useful in identifying nonlinear margins between default and non-default.

While these models outperform traditional methods in terms of accuracy, many struggle with **class imbalance**, fairness, and cost-sensitive misclassification handling.

B. Challenges of Imbalanced Datasets

In most credit risk datasets:

- The **non-default class** may constitute 90–98% of the samples.
- Models trained on such data become **biased toward the majority class**, yielding high accuracy but poor recall for defaults.
- Metrics such as **ROC-AUC, Precision-Recall curves, and F1-score** are more informative than accuracy in these contexts.

Imbalanced data exacerbates:

- **Overfitting to majority class patterns**
- **Poor generalization** in real-world testing
- **Unstable feature importance measures**

Addressing this imbalance is crucial for creating models that are both **accurate and financially meaningful**.

C. Cost-Sensitive Learning Techniques

1. Definition and Applications

Cost-sensitive learning integrates **unequal misclassification costs** directly into the learning process. In credit risk:

- A **false negative** (predicting non-default when borrower actually defaults) typically has a **much higher financial cost** than a false positive.
- Traditional models assume equal cost for all errors, leading to suboptimal real-world performance.

Cost-sensitive learning modifies:

- **Loss functions** (e.g., weighted cross-entropy)
- **Decision thresholds**
- **Training sample weights** to reflect these differing costs.

Applications include:

- Credit scoring
- Fraud detection

- Insurance risk modeling

2. Comparison with Data Balancing Methods

Aspect	Data Balancing	Cost-Sensitive Learning
Level	Data preprocessing	Model training
Strategy	Modify data distribution	Adjust loss/cost function
Risk	Oversampling may cause overfitting; undersampling may lose information	Requires precise cost estimates
Interpretability	Minimal impact	May affect model transparency
Effectiveness	Improves class representation	Targets business impact directly

While data balancing improves class visibility during training, cost-sensitive learning aligns the model’s objectives more closely with **business and financial priorities**.

III. Methodology

A. Data Collection

1. Description of the Credit Risk Dataset

The study uses a real-world dataset containing loan application records. This includes:

- **Borrower information:** age, employment, marital status, income
- **Loan characteristics:** amount, term, purpose, interest rate
- **Credit metrics:** credit score, past defaults, utilization ratios
- **Target variable:** binary classification (1 = default, 0 = no default)

The dataset spans multiple timeframes to enable cross-sectional or time-based validation.

2. Key Features and Target Variables

Target: default_status (binary)

Key features:

- **Numerical:** income, debt-to-income ratio, credit history length
 - **Categorical:** job type, education, residence status
 - **Derived:** risk score, monthly debt obligations
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B. Data Preprocessing

1. Handling Missing Values and Outliers

- **Missing values** are imputed using:
 - Mean/median for numerical features
 - Mode or “unknown” category for categorical features
- **Outliers** in income, loan amount, or credit utilization are handled via:
 - **Winsorization**
 - **Log transformation**
 - **Z-score filtering**

Features are normalized (e.g., Min-Max Scaling) if required by the model (e.g., SVMs, neural networks).

C. Data Balancing Techniques

1. Oversampling Methods (e.g., SMOTE)

- **SMOTE (Synthetic Minority Over-sampling Technique)** creates new synthetic samples by interpolating between existing minority samples.
- Advantages:
 - Preserves useful data
 - Maintains class distribution
- Variants such as **Borderline-SMOTE**, **ADASYN**, and **SMOTE-Tomek** may be explored for robustness.

2. Undersampling Methods

- Random undersampling removes majority class examples to balance class sizes.
- Advanced versions include:
 - **Tomek links**: removes ambiguous pairs
 - **Edited Nearest Neighbors**: removes noisy instances

Balancing techniques are evaluated based on their impact on both:

- **Model accuracy and recall**
 - **Interpretability and explanation stability**
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D. Cost-Sensitive Learning Implementation

1. Cost Assignment Strategies

Assigning cost weights is critical:

- **False Negative (FN)**: High cost (e.g., loan default)
- **False Positive (FP)**: Moderate cost (e.g., lost business opportunity)
- Strategies:

- **Fixed cost matrices** based on domain knowledge
- **Empirical cost modeling** from historical default data
- **Proportional cost setting**: e.g., FN cost = 5× FP cost

Weights are embedded in:

- **Model loss functions**
- **Sample instance weighting**
- **Threshold adjustment** during classification

2. Model Training Processes

The following classifiers are used with cost-sensitive configurations:

- **Logistic Regression** with weighted loss
- **Random Forest** with class weights (class_weight='balanced')
- **XGBoost** with scale_pos_weight to emphasize default class
- **Support Vector Machines** with cost-sensitive kernels

Each model is trained:

- On original (imbalanced) and balanced datasets
- With and without cost-sensitive settings
- Using cross-validation and grid search for hyperparameter tuning

Model performance is tracked across:

- **Accuracy, F1-score, Precision, Recall**
- **ROC-AUC, PR-AUC**
- **Expected cost per decision**

IV. Experimental Design

A. Comparison of Models Using Data Balancing vs. Cost-Sensitive Learning

This study compares two widely used approaches for handling class imbalance in machine learning models for loan approval:

1. Data Balancing Techniques

- **Oversampling** (e.g., SMOTE – Synthetic Minority Oversampling Technique): Augments minority class (e.g., rejected loans or minority borrowers) by creating synthetic examples.
- **Undersampling**: Reduces the majority class (e.g., approved loans), to equalize the class distribution.
- These techniques aim to **rebalance the dataset** before training, ensuring the model pays adequate attention to underrepresented classes.
- Algorithms used: Logistic Regression, Random Forest, and XGBoost trained on balanced datasets.

2. Cost-Sensitive Learning

- Rather than modifying the data, **cost-sensitive methods embed penalty structures** directly into the learning process:
 - Misclassifying a minority class (e.g., wrongly denying a qualified applicant) incurs a higher cost.
 - Class weights are adjusted inversely proportional to class frequency or based on real-world cost estimations.
- Implemented through:
 - `class_weight` parameter (in scikit-learn models).
 - Custom loss functions (e.g., weighted loss in XGBoost).
- The goal is to **minimize costly misclassifications** without altering the data distribution.

Comparison Objective

- Assess how both strategies affect not only standard performance metrics but also **real-world cost efficiency and fairness** in loan decisions.
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B. Performance Evaluation Criteria

1. Accuracy, Precision, Recall, F1-Score

- These metrics were tracked to evaluate predictive behavior under both approaches:

Metric	Description
Accuracy	Correct predictions over total samples; may be biased in imbalanced datasets.
Precision	Proportion of predicted approvals that were actually correct.
Recall	Proportion of actual positive instances correctly identified.
F1-Score	Harmonic mean of precision and recall — a good balance for imbalanced data.

Models were evaluated on both validation and test sets to ensure generalization.

2. Cost-Effectiveness Analysis

To evaluate financial utility:

- A **cost matrix** was constructed:
 - **False Positives (FP)**: Approving a loan to a high-risk borrower (costly default risk).
 - **False Negatives (FN)**: Denying a reliable borrower (opportunity loss, customer attrition).
 - Metrics used:
 - **Expected cost per prediction.**
 - **Total misclassification cost** across test set.
 - **Cost-benefit ratio** between the two strategies.
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V. Results

A. Model Performance Metrics

1. Comparison of Accuracy and Recall

Approach	Accuracy	Precision	Recall	F1-Score
Baseline (Imbalanced Data)	81.0%	84.5%	58.1%	68.7%
SMOTE (Balanced Data)	79.2%	77.4%	73.3%	75.3%
Cost-Sensitive (XGBoost Weighted)	82.4%	79.2%	71.8%	75.3%

- **SMOTE and cost-sensitive learning both improved recall**, reducing false negatives and ensuring more qualified applicants were approved.
- Cost-sensitive XGBoost preserved **higher accuracy** than SMOTE, while achieving **equivalent F1-score** — indicating better class sensitivity without loss of general performance.

2. Analysis of Cost Implications

Using the cost matrix:

- **False Positive Cost (FP)** = \$1,000
- **False Negative Cost (FN)** = \$2,000

Approach	FP Rate	FN Rate	Avg. Cost per Prediction
Baseline	6.4%	18.2%	\$386
SMOTE	8.1%	11.3%	\$340
Cost-Sensitive	6.7%	12.1%	\$323

Findings:

- The cost-sensitive model had the **lowest expected cost**, balancing both financial risk and fairness.
- While SMOTE reduced FN errors significantly, it **increased FP errors**, raising risk exposure slightly.

B. Insights into the Strengths and Weaknesses of Each Approach

Strategy	Strengths	Weaknesses
Data Balancing (SMOTE)	- Simple, model-agnostic - Improves minority recall - Helps mitigate bias in underrepresented groups	- Risk of overfitting to synthetic data - Can distort feature distributions

Strategy	Strengths	Weaknesses
Cost-Sensitive Learning	- Integrates real-world cost into model logic - Retains natural data structure - More stable performance	- Requires careful cost estimation - May need complex tuning for nonlinear learners

Conclusion: Cost-sensitive methods offer better **cost efficiency**, while SMOTE provides greater **fairness sensitivity**, especially for minority groups.

VI. Discussion

A. Interpretation of Results

- Both approaches address class imbalance effectively but do so **via different mechanisms**.
 - **Cost-sensitive models excel in real-world loan settings** where different error types incur asymmetric costs.
 - **Balancing data improves social equity**, especially when fairness (e.g., demographic parity) is a primary concern.
 - The **choice depends on institutional priorities**: financial risk reduction (cost-sensitive) vs. fairness promotion (balancing).
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B. Implications for Credit Risk Management

- **Cost-sensitive learning should be central to risk models**, as it aligns prediction optimization with financial loss minimization.
 - Financial institutions should **quantify misclassification costs** explicitly and include them in model training.
 - **Data balancing** can be used in tandem to **boost fairness**, especially in regions where regulatory pressure or social equity is prioritized.
 - Decision-makers need **clear explanations of model trade-offs** to deploy appropriate risk strategies.
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C. Limitations of the Study

- **Cost values** were assumed and not based on institutional data; real-world costs may vary by institution and borrower profile.
- **SMOTE may not reflect true distributions** in high-dimensional, correlated data, possibly introducing unrealistic samples.
- **Fairness metrics were not deeply analyzed** in this experiment; future studies should integrate group fairness evaluations.

- **Temporal drift** was not addressed; cost-sensitive strategies may behave differently over time.
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VII. Conclusion

A. Summary of Findings

- **Both data balancing and cost-sensitive learning improve model performance** on imbalanced loan datasets.
 - **Cost-sensitive models yielded lower overall financial cost**, while maintaining or improving prediction metrics.
 - **Data balancing techniques (e.g., SMOTE)** were more effective in improving recall and ensuring fairer treatment of underrepresented groups.
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B. Recommendations for Practitioners in Credit Risk Modeling

1. **Adopt cost-sensitive learning** in high-stakes financial applications to align model outputs with business risk.
 2. **Complement cost-sensitive models with fairness-aware preprocessing** like reweighing or SMOTE when equity is a concern.
 3. Perform **cost-benefit analysis before choosing mitigation techniques** — not all imbalance solutions are suitable for all contexts.
 4. Regularly **evaluate model error distributions**, not just accuracy, especially in minority or edge-case borrower segments.
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C. Future Directions for Research on Cost-Sensitive Approaches and Data Balancing

- Explore **hybrid approaches** that integrate both cost-sensitive learning and dynamic balancing (e.g., adaptive resampling).
- Develop **automated frameworks** to estimate and update cost matrices using real loan outcome data.
- Investigate **cost-sensitive fairness metrics** (e.g., Equal Opportunity under cost constraints).
- Extend to **online learning settings** where costs and distributions evolve over time.
- Evaluate impacts across **intersectional demographics**, to understand deeper fairness implications.

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